Chapter 2: Ideal Transmission Line Fundamentals

1 Ideal Transmission Line Fundamentals

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1.1 2.1 Transmission Line Structures on PCB / MCM *(p.3)*

Two common planar TX line geometries:

Type	Location	Ground
Microstrip	Outer (surface) layer	Single ground plane below
Stripline	Inner layer	Ground/power planes above AND below

- Typical conductor: copper
- Typical dielectric: **FR4** (epoxy-glass composite, $\varepsilon_r \approx 44.5$)
- For AC signals: the power plane acts as an **AC ground**.

1.2 2.2 Wave Propagation Concept *(p.4)*

Wave propagation on a TX line is analogous to a waterfront: - **Voltage** height of the water - **Current** flow of the water

This model is valid when the **rise/fall time is small** compared to the propagation delay down the trace.

1.3 2.3 TX Line Equivalent Circuit *(p.6)*

A TX line is modeled as a distributed RLCG ladder network per unit length (TEM mode assumed):

- **R** (/m): losses from finite conductor conductivity
- **L** (H/m): energy stored in the magnetic field
- **C** (F/m): energy stored in the electric field
- **G** (S/m): losses from finite dielectric resistance

TX line RLCG equivalent circuit

TEM (Transverse Electromagnetic Mode): no E or H field components propagate in the z-direction (propagation direction).

1.4 2.4 Characteristic Impedance *(pp.79)*

1.4.1 General (lossy) formula:

$$Z_0 = \sqrt{\frac{Z}{Y}} = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

(chap2.pdf, p.9)

1.4.2 Lossless approximation (valid for most practical TX lines):

$$Z_0 = \sqrt{\frac{L}{C}}$$

(chap2.pdf, p.9)
R and G only matter at very high frequencies or very lossy lines.

1.4.3 Microstrip approximation:

$$Z_0 \approx \frac{87}{\sqrt{\varepsilon_r + 1.41}} \ln\left(\frac{5.98H}{0.8W + T}\right) \quad \text{Ohms}$$

(chap2.pdf, p.9)
Valid when $0.1 < W/H < 2.0$ and $1 < \varepsilon_r < 15$. (W = trace width, H = height above ground, T = trace thickness)

1.4.4 Symmetric Stripline approximation:

$$Z_{0,\text{sym}} \approx \frac{60}{\sqrt{\varepsilon_r}} \ln\left(\frac{4H}{0.67\pi(T + 0.8W)}\right) \quad \text{Ohms}$$

(chap2.pdf, p.10)
Valid when $W/H < 0.35$ and $T/H < 0.25$.

For **offset stripline** (trace not centered between planes), use the harmonic mean of two symmetric stripline values at heights A and B:

$$Z_{0,\text{offset}} \approx 2 \cdot \frac{Z_{0,\text{sym}}(2A, W, T, \varepsilon_r) \cdot Z_{0,\text{sym}}(2B, W, T, \varepsilon_r)}{Z_{0,\text{sym}}(2A, W, T, \varepsilon_r) + Z_{0,\text{sym}}(2B, W, T, \varepsilon_r)}$$

(chap2.pdf, p.10)

1.5 2.5 Propagation Velocity and Time Delay *(pp.1112)*

$$v = \frac{c}{\sqrt{\varepsilon_r}}$$

$$\text{PD} = \frac{1}{v} = \frac{\sqrt{\varepsilon_r}}{c}$$

$$\text{TD} = \frac{x\sqrt{\varepsilon_r}}{c}$$

(chap2.pdf, p.11)
where $c = 3 \times 10^8$ m/s, x = line length, ε_r = dielectric constant.
Alternate form using distributed parameters:

$$\text{TD} = \sqrt{LC}, \quad \text{PD} = \sqrt{L_{\text{unit}} C_{\text{unit}}}$$

(chap2.pdf, p.11)

1.5.1 Effective dielectric constant for microstrip:

$$\varepsilon_e = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left(1 + \frac{12H}{W}\right)^{-1/2} + F - 0.217(\varepsilon_r - 1)\frac{T}{\sqrt{WH}}$$

(chap2.pdf, p.12)
where:

$$F = \begin{cases} 0.02(\varepsilon_r - 1)\left(1 - \frac{W}{H}\right)^2 & \text{for } W/H < 1 \\ 0 & \text{for } W/H > 1 \end{cases}$$

Comparison: On the same FR4 board, stripline has a **larger** effective ε_r (slower, larger TD), while microstrip has a **smaller** effective ε_r (faster, smaller TD).

1.6 2.6 SPICE Equivalent Circuit Models *(pp.1314)*

A TX line cannot be modeled with infinite elements; use enough segments based on minimum rise/fall time:

$$\text{segments} \geq 10\left(\frac{x/v}{T_r}\right) = 10\left(\frac{\text{TD}}{T_r}\right)$$

(chap2.pdf, p.14)

Circuit parameters per segment:

$$C_{\text{seg}} = \frac{x \cdot C/\text{meter}}{\text{segments}}, \quad L_{\text{seg}} = \frac{x \cdot L/\text{meter}}{\text{segments}}$$

$$\text{TD}_{\text{seg}} = \sqrt{L_{\text{seg}} C_{\text{seg}}} \leq \frac{T_r}{10}$$

(chap2.pdf, p.14)

1.6.1 Example 2.1 Creating a TX Line Model *(pp.1516)*

Given: $Z_0 = 50\,\Omega$, $x = 5$ in., $\tau_r = 2.5$ ns, $\varepsilon_r = 4.5$, stripline geometry ($W = 5$ mil, $T = 0.7$ mil, $H = 14.7$ mil).

$$\text{TD} = 5 \text{ in.} \times 0.0254 \text{ m/in.} \times \frac{\sqrt{4.5}}{3 \times 10^8 \text{ m/s}} = 898 \text{ ps}$$

$$L_{\text{total}} = \text{TD} \times Z_0 = 898 \times 10^{-12} \times 50 = 44.9 \text{ nH}$$

$$C_{\text{total}} = \frac{\text{TD}}{Z_0} = \frac{898 \times 10^{-12}}{50} = 17.9 \text{ pF}$$

Segments needed: $10 \times \frac{5 \text{ in.} \times 0.0254}{2.5 \text{ ns} \times 1.41 \times 10^8} = 3.6 \rightarrow$ use **4 segments**

$$C_{\text{seg}} = 17.9/4 = 4.48 \text{ pF}, \quad L_{\text{seg}} = 44.9/4 = 11.23 \text{ nH}$$

(chap2.pdf, pp.1516)

1.7 2.7 Launching an Initial Wave *(p.17)*

When a source V_s with source impedance Z_s drives a TX line of impedance Z_0, the initial (incident) voltage is set by a voltage divider:

$$V_i = V_s \frac{Z_0}{Z_0 + Z_s}$$

(chap2.pdf, p.17)

When $Z_L = Z_0$ (matched): V_i is the DC steady-state value no reflections.

1.8 2.8 Reflection Coefficient *(p.18)*

At any **junction** (impedance discontinuity):

$$\rho = \frac{V_{\text{reflected}}}{V_{\text{incident}}} = \frac{Z_t - Z_0}{Z_t + Z_0}$$

(chap2.pdf, p.18)

Special cases: - **Matched** ($Z_t = Z_0$): $\rho = 0$ - **Short** ($Z_t = 0$): $\rho = -1$ - **Open** ($Z_t = \infty$): $\rho = +1$

Reflections continue bouncing back and forth (counter-reflections) until steady state is reached.

1.9 2.9 Lattice (Bounce) Diagrams *(pp.2021)*

A bounce diagram tracks wave amplitude vs. time: - Diagonal lines = waves traveling on the line - Time increases downward - Each time step = TD (one-way propagation delay) - At each end, multiply by the reflection coefficient at that end

Overdriven ($Z_0 > Z_s$): initial $V_i > V_{ss}$ ringing above steady state.

Underdriven ($Z_0 < Z_s$): initial $V_i < V_{ss}$ signal rises gradually to steady state.

Rise time effect: if $\tau_r > 2\text{TD}$, reflections arrive during the rise transition waveform distortion (ledges, glitches) occur. *(p.27)*

1.10 2.10 Reflections from Reactive Loads *(pp.2830)*

1.10.1 Capacitive load (no loss):

- Capacitor initially looks like a **short** ($\rho = -1$), later like an **open** ($\rho = +1$).
- Adding C_L slows down the data rate; signal at load has a time constant dependent on C_L.

1.10.2 Capacitive load (lossy parallel $R_L C_L$):

With $Z_s = Z_0$, time constant $= C_L \times (R_L \| Z_0)$. Adding R_L makes τ smaller (faster settling). *(p.29)*

1.10.3 Inductive load:

- Inductor initially looks like an **open** ($\rho = +1$), then discharges with time constant $\tau = L/Z_0$.
- Produces a spike at $t = 2\text{TD}$. *(p.30)*

1.11 2.11 Termination Schemes *(pp.3136)*

1.11.1 Goal: match the impedance to eliminate reflections.

Method	How	Pros	Cons
On-die source	I-V curve of output buffer Z_0	No extra components	Difficult to achieve; varies with process/voltage/temp
Series source	Add R_s in series so $R_s + Z_{\text{driver}} = Z_0$	Small total variation	Extra resistor cost/area
Load (parallel)	$R_L = Z_0$ at load end	Low-impedance buffers OK	DC current draw power/thermal issues
AC load	Series $R = Z_0$, C_L at load; $RC \approx 12\tau_r$	No DC dissipation	Increases signal delay; extra components

AC termination: $R = Z_0$, $RC_L = 1 \sim 2\tau_r$ *(p.35)*

Line length guidelines *(p.36)*: - **Short lines** source termination preferred (preserve DC levels) - **Long lines** load termination preferred (prevent reflections traveling back)

1.12 2.12 Bergeron Diagrams *(pp.2526)*

Used when nonlinear sources/loads are present. Solve graphically: 1. Plot the source I-V line (slope $= -1/Z_0$, passing through $(V_s, 0)$). 2. Plot the load I-V curve. 3. Intersection = operating point of the incident wave. 4. Repeat for reflected waves.

1.13 2.13 Additional Example 50 Microstrip Design *(pp.3743)*

Given: $Z_s = 30\,\Omega$, edge rate = 100 ps, swing 02 V, $Z_0 = 50\,\Omega$, $x = 5$ in., $\varepsilon_r = 4.0$, microstrip.

Key results: - Trace geometry: $T = 1.0$ mil, $W = 5$ mil, $H = 3.2$ mil (approximated from Z_0 formula) - ε_e: since $W/H > 1$, $F = 0$ - Propagation velocity computed from ε_e, then TD $= x/v$ - Minimum 72 segments required - L and C per segment computed from total $L_{\text{total}} = \text{TD} \times Z_0$ and $C_{\text{total}} = \text{TD}/Z_0$

(chap2.pdf, pp.3943)